

Application of household disruption data to delineate critical infrastructure resilience characteristics in the aftermath of disaster: A case study of Bhaktapur, Nepal

Hadi Nazarnia^{a,*}, Hadi Sarmasti^b, W. Olivia Wills^c

^a Moss School of Construction, Infrastructure and Sustainability, Civil and Environmental Engineering Department, Florida International University, Miami, FL 33174, USA

^b Department of Civil Engineering, Sahand University of Technology, Tabriz, Iran

^c Department of Biological Sciences, Florida International University, Miami, USA

ARTICLE INFO

Keywords:

Resilient city
Disaster resilience
Disruption data
Civil infrastructure systems
Emergency response
2015 Gorkha Earthquake

ABSTRACT

Assessing the resilience and interdependence of civil infrastructure systems in order to provide a resilient community for resident has become a prevalent topic among policy-makers, authorities, and scholars around the world. Despite significant advances in this field, there is still a lack of research regarding comprehension of the actual behavior of civil infrastructures in response to hazards and disasters due to their complexity and hidden interconnectivity.

This study utilizes disruption data and a network analysis framework to describe resilience dimensions in two infrastructures and further validate the previously presented model. Following the evaluation of infrastructures and the previous model, interdependency between two lifelines will be discussed. Service disruption data is a valuable resource to confirm various theoretical methods. A case study of lifelines in Bhaktapur, Nepal is presented to show applicability of the network analysis model and service disruption data in evaluating the interdependencies between these two lifelines. The resilient characteristics of water infrastructure and the interdependence to power infrastructure are explained.

The contribution of this article is threefold: (i) validation of a novel network approach using disruption information, (ii) analysis of recoverability, sensitivity, and types of disruption patterns, and (iii) comparison of behavior and interdependencies between the infrastructures.

1. Introduction

Resilience of Critical infrastructures (CI) play a vital role in providing reliable service to the communities. Everyday life relies on the proper functioning of various CISs, such as electric power, water and gas distribution, and transportation infrastructures (Blokus, 2018; Kröger, 2008; Kröger and Zio, 2011). Resilience of these life lines bonds with well understanding the behavior of them in the aftermath of any natural or man-made disasters. In another word, to have a disaster resilient community, mainly in a time of extreme events and post disaster periods, characterizing resilience and interdependencies between infrastructures should be conceived (Madni and Jackson, 2009). In 1998, US president's Commission on Critical Infrastructure Protection (CCIP) officially declared that the trusty performance of complex interdependency of infrastructure leads to safe, prosper economic, and ensure welfare of the society (Robert et al., 2002; Procyk and Dhariwal, 2010; Gillette et al., 2002).

Nowadays, although most of the authorities and scholars realize the importance of these infrastructure systems in supporting everyday activities, the resilience, interconnectedness, and dependencies of the lifelines in urban areas have become a big challenge for decision makers and planners. Understanding the behavior of the infrastructure after catastrophic events is challenging (Blokus, 2018). In addition, recoverability dimension in resilience metrics becomes an emerging topic in the aftermath of disasters.

Due to a gap of knowledge in interdependency among urban infrastructures, damage propagation, and recoverability in one-side (Goto et al., 2008), and different context of resilience in developing country due to social, economic, technological and political aspects in other side, the objective of this paper is to use service disruption data to characterize infrastructure resilience, better understanding the behavior of the systems, and also analyze infrastructure interdependencies. To this end, the case study of 2015 Nepalese Earthquake, by concentration on the Water and Electric infrastructures, has been

* Corresponding author.

E-mail address: hnaza001@fiu.edu (H. Nazarnia).

<https://doi.org/10.1016/j.ssci.2019.02.014>

Received 16 June 2018; Received in revised form 2 February 2019; Accepted 12 February 2019

Available online 20 February 2019

0925-7535/ © 2019 Elsevier Ltd. All rights reserved.

investigated to understand how the damages propagated and how these dependencies affect the functionality of other infrastructures. Accordingly, disruption service data and status of end-users is used to assess the resilience of infrastructure.

A novel framework (Nazarnia and Sarmasti, 2018) is applied to water infrastructure service provided by the government in Bhaktapur, Nepal and water and power infrastructures' disruption data is used to evaluate and confirm existing interdependencies between these two lifelines. This disruption based model is able to capture the characteristics of the lifelines after a devastating earthquake in Nepal without detailed information from the physical network damages, which might be inaccessible in most extends, hence, it resolves the corresponding obstacles in data collection to create a network.

Results in the infrastructure assessment demonstrate sensitivity, local versus system-wide damages, and recoverability of the system. Therefore, presented network analysis framework is able to characterize infrastructure resilience and disruption patterns from customers' service status data.

2. Background

The societal need for uninterrupted performance of CISs has resulted in a significant surge in the number of publications on the resilience of CISs. However, despite substantial advances, assessing the resilience of CISs remains challenging. CISs are composed of visible and invisible elements in addition to parameters which cause complexity in the network. Such elements and parameters create interdependencies and complex socio-economic parameters, as well as several other influential parameters, in the function of the civil infrastructures (Gay and Sinha, 2013).

Improved understanding of resilience is highly significant for complex systems. Scholars have utilized several approaches to study the resilience of the CISs, all of which can be divided into two broad categories of qualitative and quantitative resilience assessment (Gao et al., 2016; Zio et al., 2008; Holmgren, 2006; Hosseini et al., 2016). Among the quantitative approaches, one robust method is the network analysis approach. There are many reasons to justify the application of network analysis in the infrastructure system assessment. First of all, the network has been used to represent many man-made, social, and biological systems, such as transportation infrastructures and underground systems. In terms of urban studies, the network approach has been applied to evaluate the traffic flow and land use changes (Porta et al., 2006). Secondly, infrastructure networks are considered to be supply networks, the resilience of which is defined as the potential for remaining operational under any disruption and preserving its connectedness despite structural or functional losses (Zhao et al., 2011). That is why the analysis of the infrastructure resilience using different methods of network analysis is inevitable. These reasons, as well as several other studies on the assessment of infrastructural systems, establish that the network method can be considered a strong tool in the study of CISs (Zio et al., 2008; Hosseini et al., 2016; Masucci et al., 2009; Bompard et al., 2009; Crucitti et al., 2005; Yazdani and Jeffrey, 2011; Yazdani and Jeffrey, 2012; Dueñas-Osorio et al., 2007).

Researchers concur that networks can exhibit the interaction of systems, and such networks can be used to monitor, and control systems (Liu et al., 2016; Reggiani et al., 2015). Therefore, the network approach is a powerful tool in better understanding the resilience of CISs. Generally, networks are shown as a graph in mathematical format and graph theory, as a comprehensive mathematical tool, has been used to investigate the characterizations of the infrastructure system and human interactions (Clauaset et al., 2008). For example, Chai et al. (2011) highlighted the vulnerability of various infrastructures with principles of the social network analysis. The scholars remarked that the interdependency of the infrastructures is similar to the interaction of the individuals in society and prioritized the oil and gas infrastructure as the most important infrastructure, which led to the

cascading failure of the entire system (Chai et al., 2011). Eisenberg et al. (2017) combined social network analysis and power grid network analysis in South Korea to provide new insight regarding organizational inclusion with blackout risk. The researchers utilized various measures of network analysis (Eisenberg et al., 2017). Yazdani and Jeffrey (2011) developed a water distribution system as a graph with nodes and physical links, and used the graph to examine several resiliency components; namely reliability, efficiency, and robustness. To quantify the resilience of the water infrastructure network, some of the network theory metrics had been used, not only to evaluate the existing network, but also to increase the redundancy of water networks in a time of network expansion (Yazdani and Jeffrey, 2011; Yazdani and Jeffrey, 2012; Kaya et al., 2018). Notably, simplifications of networks on these papers were inevitable (Didier et al., 2017). Furthermore, there are numerous parameters affecting the resilience of CISs. For example, human-infrastructure coupling or interdependency of the civil infrastructure (Nazarnia et al., 2016). Correspondingly, the disruption status of the end-users has not received much attention and the authors think studying disruption status may be a new tool on characterizing the resilience of the community. In this paper, the measure of clustering in network analysis has been used to characterize the resilience of the CISs in the aftermath of the Nepalese earthquake.

As mentioned before, graph theory is a beneficial tool for understanding network resilience; however, the application of a number of hybrid approaches can be extremely fruitful in this matter. For instance, Winkler et al. (2011) presented a hybrid assessment methodology to evaluate the interdependencies among the infrastructures. They connect the physical fragility model with the topology of the actual connections between infrastructures using some of the network measures, such as betweenness, clustering, node degree, and Euclidean distance to assess the failure propagation in the networks. Others have coupled network-based methods with the structural reliability theory to evaluate the seismic performance of infrastructure systems (Winkler et al., 2011). Although all of these methods have shed more light over the resilience of civil infrastructures, the methods require detailed information to create an associated network (Blokus, 2018). In this paper, the characteristics of the disruption and resilience of the CISs will be evaluated with regard to a few measures of the social network topology. The proposed framework borrows several concepts from sociology (i.e., peer) to form the network from the service status of end users in the system. These components of the models create a network similar to a social network. The resilience of networks is evaluated in a variety of ways, including network performance and connectivity analysis. (Gao et al., 2016) characterize the/a system's state with the average nearest-neighbor activity and discuss the system's state with outgoing and incoming weighted degree (Kröger and Zio, 2011). By utilizing the measures of network analysis such as clustering coefficient, the designed network provides a number of important measures and elements of resilience, capacity of response, and type of perturbation in the system in a CIS. Other measures of network analysis, such as density, degree centrality, or modularity, are beyond the scope of this paper and need more contemplation; these measures may provide more insight relevant to the resiliency of CISs (Nazarnia and Sarmasti, 2018).

3. Materials and methods

To study infrastructure network resiliency, various data regarding failures in the network components and perturbations to the civil infrastructure is required; for instance, the spatial information of the damages, type of the damaged and undamaged components, and the connectivity of the components (topology of the network). This required information to conduct a topology-based analysis of infrastructure networks is difficult to acquire in most cases because in the post disaster response period, time is imperative to saving live of inhabitants and a major concern is to rescue the injured in a timely manner (Blokus, 2018). In addition, the outcomes of topology-based

models mainly provide insights regarding the vulnerability of networks. Modeling also enables the identification of critical nodes in a network. Hence, topology-based network models are more useful in the evaluation of the serviceability of infrastructure networks prior to any disasters. For a post-disaster study, collecting data regarding disruption in infrastructure services provides critical information. Most of the existing studies utilize the extent of service disruptions (e.g., the number of households that experienced electricity disruption) and the speed of restoration (i.e., the duration of service disruptions). Additional insights can be gained by studying disruptions in infrastructure networks; however, the lack of methodological approaches has hindered the analysis of disruption patterns in this area.

To address this gap and provide a means for better understanding disruptions in infrastructure networks, this study uses a novel methodology based on the theoretical underpinnings of social network analysis and uses the feature of network topology and measures of network analysis. According to social network theories, two nodes in a network are considered to be “peers” if: the nodes have a similar status, have social attributes, or communicate and work with each other. Based on the same premise from the social network, in the proposed methodology infrastructure users can be considered as “peers” if they have a similar status in terms of access to infrastructure services. Based on this fundamental premise, infrastructure users are considered as nodes and their connectivity is modeled with links according to the status of their infrastructure services and disruptions at different points in time in the aftermath of a disaster. Then, through the analysis of the structure of the social network of infrastructure users, disruption patterns and the behavior of infrastructure networks are examined. The proposed methodology is capable of providing insights about a network's sensitivity to disruptions, service disruption patterns, and the capacity of the infrastructure network to respond without the need for information related to age and dependencies of physical infrastructure.

Applied methodology for the assessment of service disruptions in infrastructure networks is composed of five steps:

1. Determine the infrastructure service state.
2. Create the adjacency matrix for social network at different time intervals.
3. Model the networks at different time intervals (social network).
4. Examine the characteristics of infrastructure disruptions.
5. Conduct the trend analysis over time in social network (Analysis of the changes).

All these steps of framework to create networks are provided in (Nazarnia and Sarmasti, 2018).

According to (Nazarnia and Sarmasti, 2018), the presented methodology is resulted to a figure for clustering coefficient versus time. In this figure, the distance from the vertical axis demonstrates the ability of the infrastructure network to recover disconnected parts. In other words, the distance represents the time needed to bounce back to the nearly normal serviceability status. Less distance from the time origin (i.e., disruptive event occurrence) indicates a high capacity of response and a more resilient system. Furthermore, lower clustering coefficient at the time of disaster occurrence is equal to less sensitivity and being more resilient and show internal inertia of the system to damp the disruptive event (Nazarnia and Sarmasti, 2018).

4. Data collection and case study

4.1. Nepal-Bhaktapur

The “7.8-magnitude” earthquake on April 25, 2015 and several aftershocks struck the capital of Nepal, Kathmandu, and the surrounding areas. The infrastructures and buildings were severely damaged or destroyed and around 3.86 billion USD worth of assets were damaged. Based on the preliminary assessment, perturbations to

infrastructures were estimated to be about 50 billion NRP (approximately 0.47 Billion USD). This devastating earthquake led to many electric power cuts, frequent disruptions in the water system, and road blockages in the affected regions (Nazarnia et al., 2016; Mostafavi et al., 2017). The Bhaktapur Municipality with an area of 6884 km², is one of the greatest metropolitan area in the eastern part of the Kathmandu valley (Nazarnia and Sarmasti, 2018).

The development of urban areas is continuously increasing, which changes the complexity of the behavior of the infrastructures in a time of disaster. Kathmandu valley is one of the most populated areas near the epicenter of the Gorkha 2015 earthquake. According to the studies, before the earthquake in 2015, the expectation was that Nepal would be extremely vulnerable to an earthquake due to rapid urbanization, lack of appropriate management and supervision in the development of the urban area, and poor maintenance and aging of infrastructures (Rasoulkhani et al., 2017; Dixit, 2014).

4.2. Domestic water service infrastructure

A significant increase in the population of this area, with a population of 81,747 (2011 population census), and insufficient water sources had created various difficulties for water supply agencies even prior to the disaster. Despite financial limitation, the government of Nepal has prioritized improving clear water shortage in the area. However, water supply facilities had been in poor condition and were not operating properly.

Fig. 1 shows the water supply and water distribution network in Bhaktapur; consisting of 50 km total length distribution pipeline.

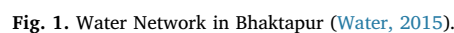
Old water conveyance, urban electricity dependent treatment plants (i.e. Bansbari Water Treatment Plant), and a gravity flow system conveyance system without an overhead tank are the characteristics of the water infrastructure in Bhaktapur. The report shows limited service hours for water service and a huge gap between water supply and demand. This shortage is compensated with tanker supply water and/or well water. Many households draw water by private pumps which store the water in tanks and create low pressure water. All of these constraints were an incentive to use disruption data to study the resilience of the lifeline in Nepal. The data used in this study is from water provided by the government and other methods of water supply are eliminated for the purpose of this paper. In order to eliminate different external factors and physical topology of the water network, disruption data from the actual event would be a beneficial resource to assess the resilience of a system (Bhaktapur Water Supply, 2015).

4.3. Electric power infrastructure

The power grid network comprises different elements, such as transmission lines with different voltages (e.g., 11, 66, 132 kV), generator substations, and distribution substations, which receive and transmit power to the other substations or distribute power in local distribution grids. Of course, in a time of disaster, any damages to the components of the electric power infrastructure lead to different patterns of disruption in the system. For instance, perturbation to the generator substations of the city give rise to a blackout in the entire region and when the generator is fixed, large regions recover service instantly. Conversely, damages in the distribution grids (e.g., falling poles) may result in the local perturbation (Crucitti et al., 2004; Mostafavi et al., 2017).

Assessment of CI performance after any disruptive event have a significant role in recoverability of an infrastructure (Blokus, 2018). In addition, studying these disruptions in the aftermath of the earthquake not only provides a novel sight in the assessment of the characteristics and behavior of the CIs, but can also be used to verify other theoretical methods.

Various approaches have been used to collect the data related to the infrastructures in the regions. Data in this paper has been collected from



Data was obtained from field surveys to monitor in detail status of the water supply and electricity in the households. Initially, the number of interviewed households was 223. Since different customers have different orders of weight, compiled data considers only water services provided by the government to the private sector which have a same order of weight in resilient assessment. Therefore, the importance of services or the client rank is the same in the used data. In addition, households with improper information relating to power outage have been eliminated to take into account interdependencies between water and electric power infrastructures and how service disruption can

These data points are utilized to validate the output of the novel model. In these samplings, a number of questions were asked about the damages to different infrastructures (such as water, electric power, and roads). The interviews provide useful data about the duration of the infrastructure service disruptions in these households. Among the different infrastructure alternatives for the users, the electric power infrastructure and water supply system are selected to verify the disruption-based model.

Figs. 3 and 4 show the different networks for electric power and water lifelines in the aftermath of the 2015 Gorkha earthquake in sequential order from immediately after the earthquake to approximately 180 days later. The comparison between developed networks demonstrates recoverability of the disruptive event in the specific infrastructure system. Initially, most of the households experienced disruption in service, resulting in a very dense network. Accordingly, this network has a higher clustering coefficient compared to the subsequent networks. With the passage of time after the earthquake, the number of edges in the networks decreases gradually, which implies the households regain their electricity/water service; therefore, the associated

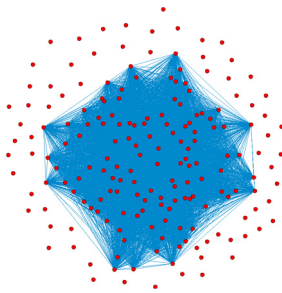


Fig. 2. Participation of households with disruption in the service to map a network.

clustering coefficient of each network decreases. These networks have been created in the following days based on the interview forms: day 1, 3, 6, 14, 28, 84, 168, and 180.

In the next stage, all the cluster coefficients retrieved from the different networks are graphed over time of response. Fig. 5 shows the trend of variation in clustering coefficient in the electric lifeline. The required steps to create the graph are: to identify similarity in the disruption state of the households (i.e., “peer”), obtain the adjacency matrices, develop different networks in a time division, obtain associated clustering coefficients, and then plot the coefficients versus time.

As shown in Fig. 6, water infrastructure in Bhaktapur demonstrates different behavior compared to the electric lifeline. A lower clustering coefficient closer to the more immediate days after the earthquake, the more resilient a system is. This characteristic of water infrastructure in Bhaktapur, is consistent with other studies on water lifelines in Nepal and confirm accuracy of the methodology and assessment (Mostafavi et al., 2017).

Three characterized results are retrieved from the model for the case study: (1) the amount of sensitivity in the beginning of the restoration process is different for each infrastructure, (2) disruption patterns (local and system-wide) of the network are mostly local for the power lifeline and system-wide for the water lifeline, and (3) capacity of response of the electric power infrastructure in Bhaktapur is low and more time was needed to restore power. In contrast, water infrastructure was recovered faster. All these characteristics are shown in Figs. 5 and 6, which convey that the electric system was more sensitive to the disruptive event. The system-wide disruption recovered within 25–30 days after the catastrophic event and led to recovery of water service. This character depicts interdependency of the water service to the electricity. Moreover, the damage in the electric power infrastructure was local and recovered slowly, whereas water infrastructure recovered faster. Although supply and demand difference in the water infrastructure adds more complexity to the system, the level of service was set based on the pre disaster situation.

Study of water infrastructure in Bhaktapur indicated that despite experiencing several earthquakes, the water distribution network was not the main reason for network disruptive and water closure for the end users. According to the data, the blackout after the earthquake created a cascading failure in the lifeline performance. Furthermore, a comparison of Figs. 5 and 6 confirms that water infrastructure is more resilient than electric power infrastructure and despite experiencing various extreme events and age of the distribution system, recoverability of a large portion of this lifeline required almost 25 days.

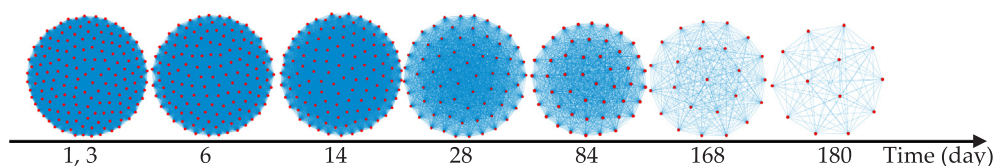


Fig. 3. Dynamic changes in the disruption-based network for electric power infrastructure at a different time from the baseline to the recovered situation (left to right) in the aftermath of the earthquake.

6. Concluding remarks

This work has highlighted the importance of service disruption data in characterizing resilience and interdependency of critical infrastructures after any extreme event. Disruption information for water and electric power infrastructures after the 2015 Gorkha earthquake has been used in this paper to verify capability of this novel approach in providing reliable insights regarding infrastructure behavior.

Based on the service disruption to the households, the semantic graph in each time frame was produced to demonstrate the mutual relationship and connectivity of the households. Since this disruption data is able to delineate various dimensions of the critical infrastructure, it has been used in the case of electric power and water lifelines in Nepal.

In the modeling process at the peer level, different unknown end users in different parts of the city are connected to each other based on similarities in disruption status (i.e., disruption period) of their service in the aftermath of a disaster. Peer notion is applied to link the disrupted households and build the proposed model. Social networks corresponding to the infrastructure in each time interval are created and analyzed. This algorithm is suited to be used in different CISs, such as water and transportation. Three main characteristics of infrastructure systems can be drawn from the analysis. First, the sensitivity of the infrastructure networks relies on the clustering of the disrupted end-users and the tendency of nodes to be clustered together. A higher clustering coefficient indicates a higher level of system sensitivity after a disruption. Second, the rapid decrease in connectivity of a dense node compared to a less dense node leads to a major decrease in clustering coefficient and demonstrates a system-wide perturbation in the entire CIS and vice versa. Third, distance from the origin in time illustrates capacity of response in the system. Resulting analysis produced by the model evaluate the patterns of disruptions in the infrastructure networks (Carley and Reminga, 2004).

Comparison of the disruption data in water and electric power services in the Gorkha 2015 earth quake implies interdependencies between the water supply system provided by the government, and the electric power. Asian Development Bank's (ADB) urban water supply and sanitation report from September 2018 suggests that providing a power backup generator ensures water supply operation in Nepal and reduces the probability of such physical interdependencies and therefore increases resilience of the water supply system (Proposed Loan Nepal, 2018).

Initial findings in the case of Bhaktapur reveal that the disruption in the water infrastructure is more system wide whereas disruption in the electric power infrastructure is more local. Results of the simulations are consistent with the other studies in Nepal (Mostafavi et al., 2017). This innovative framework opens a new chapter in the assessment of the resiliency of the communities and the future works will be able to focus on the contribution to this notion in the infrastructure systems.

Lack of prior studies in the association between the resilience of the infrastructures in the system and service status of end users was the main incentive to focus on disruptions in the infrastructure systems after a disaster. This work can be further extended in several directions. First, the proposed network approach could be enhanced through the use of existing techniques in social network analyses and development of better algorithms for assessing the resilience of infrastructures with consideration of the disruption data and patterns (Zhao et al., 2011). Second, this method advances the concept of modeling physical civil

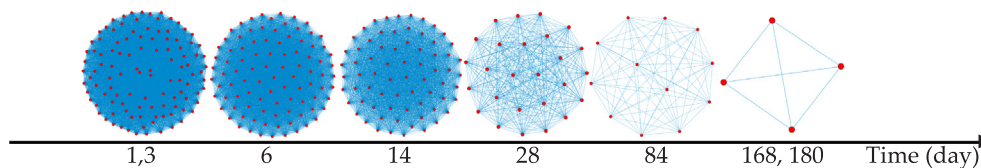


Fig. 4. Dynamic changes in the disruption-based network for water infrastructure at a different time from the baseline to the recovered situation (left to right) in the aftermath of the earthquake.

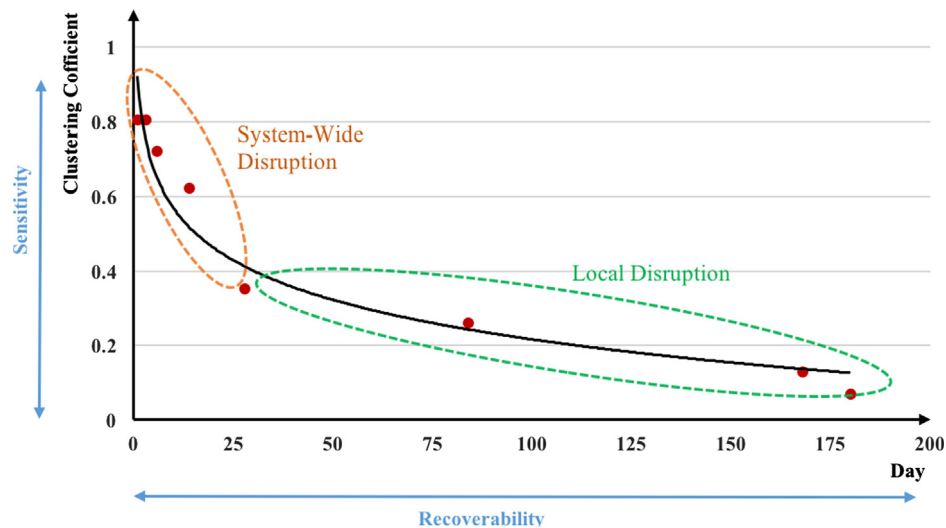


Fig. 5. Variation of the clustering coefficient in electric power infrastructure in the case study of Nepal (Bhaktapur) over time.

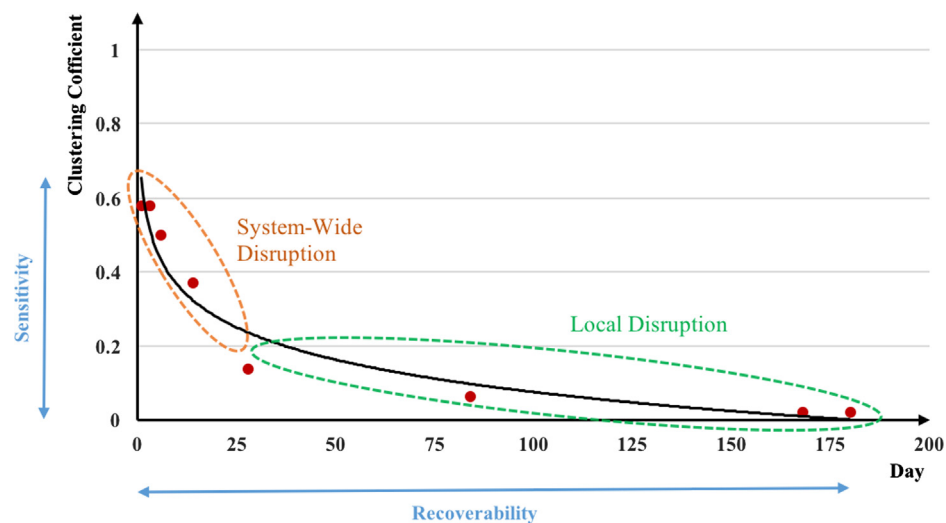


Fig. 6. Variation of the clustering coefficient in water electric infrastructure in the case study of Nepal (Bhaktapur) over time.

infrastructure networks via numerous social networks and assesses the interdependency or dependency of the infrastructures and human-infrastructure coupling (Crucitti et al., 2004; PDNA, 2015). Finally, the proposed model creates the opportunity to apply prediction methods to improve civil infrastructure networks and increase their robustness.

Acknowledgment

This material is based in part upon work supported by the National Science Foundation under Grant No. CMMI-1546738. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation. Authors would like to thank Dr. Nipesh Pradhananga and Dr. Irtyshad Ahmad for their endless support.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ssci.2019.02.014>.

References

- Balaei, B., Wilkinson, S., Potangaroa, R., Hassani, N., Alavi-Shoshtari, M., 2018. Developing a framework for measuring water supply resilience. *Nat. Hazard. Rev.* 19 (4), 04018013.
- Bhaktapur Water Supply Improvement Project in the Federal Democratic Republic of Nepal Study Report March 2015 retrieved January 2019 from: < https://www.mhlw.go.jp/file/06-Seisakujouhou-10900000-Kenkoukyoku/0000092634_7.pdf > .
- Blokus, A., 2018. Quantitative assessment of economic, social and environmental impacts of critical infrastructure disruptions. In: 2018 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), IEEE, pp. 824–828.
- Bompard, E., Napoli, R., Xue, F., 2009. Analysis of structural vulnerabilities in power transmission grids. *Int. J. Crit. Infrastruct. Protect.* 2 (1), 5–12.

- Carley, K.M., Reminga, J., 2004. Ora: Organization risk analyzer (No. CMU-ISRI-04-106). Carnegie-mellon univ Pittsburgh pa inst of software research internet.
- Chai, C.L., Liu, X., Zhang, W.J., Baber, Z., 2011. Application of social network theory to prioritizing Oil & Gas industries protection in a networked critical infrastructure system. *J. Loss Prevent. Process Industries* 24 (5), 688–694.
- Clauset, A., Moore, C., Newman, M.E., 2008. Hierarchical structure and the prediction of missing links in networks. *Nature* 453 (7191), 98.
- Crucitti, P., Latora, V., Marchiori, M., 2004. A topological analysis of the Italian electric power grid. *Physica A: Statist. Mech. Its Appl.* 338 (1), 92–97.
- Crucitti, P., Latora, V., Marchiori, M., 2005. Locating critical lines in high-voltage electrical power grids. *Fluctuat. Noise Lett.* 5 (2), L201–L208.
- Didier, M., Broccardo, M., Esposito, S., Stojadinovic, B., 2017. A compositional demand/supply framework to quantify the resilience of civil infrastructure systems (Re-CoDeS). *Sustain. Resilient Infrastruct.* 1–17.
- Dixit, A.M., 2014. Earthquake disaster risk management efforts in Nepal. In: *International Symposium Geohazards: Science, Engineering and Management*.
- Dueñas-Osorio, L., Craig, J.L., Goodno, B.J., 2007. Seismic response of critical interdependent networks. *Earthq. Eng. Struct. Dyn.* 36 (2), 285–306.
- Eisenberg, D.A., Park, J., Seager, T.P., 2017. Sociotechnical network analysis for power grid resilience in South Korea. *Complexity*, 2017.
- Gao, J., Barzel, B., Barabási, A.L., 2016. Universal resilience patterns in complex networks. *Nature* 530 (7590), 307–312.
- Gay, L.F., Sinha, S.K., 2013. Resilience of civil infrastructure systems: literature review for improved asset management. *Int. J. Crit. Infrastruct.* 9 (4), 330–350.
- Gillette, J.L., Fisher, R.E., Peerenboom, J.P., Whitfield, R.G., 2002. Analyzing water/wastewater infrastructure interdependencies (No. ANL/DIS/CP-107254). Argonne National Lab., IL (US).
- Goto, Y., Suzuki, T., Suetomi, I., Shoji, Y., Tsuruta, M., Kataoka, S., Suzuki, H., 2008. Damage propagation caused by interdependency among critical infrastructures. The 14th World Conference. on Earthquake Engineering.
- Holmgren, Å.J., 2006. Using graph models to analyze the vulnerability of electric power networks. *Risk Anal.* 26 (4), 955–969.
- Hosseini, S., Barker, K., Ramirez-Marquez, J.E., 2016. A review of definitions and measures of system resilience. *Reliab. Eng. Syst. Saf.* 145, 47–61.
- Kaushik, H., Bevington, J., Jaiswal, K., Lizundia, B., Shrestha, S., 2016. Buildings (EERI Earthquake Reconnaissance Team Report: M7. 8 Gorkha, Nepal Earthquake on April 25, 2015 and its Aftershocks) (pp. 5–1). Earthquake Engineering Research Institute.
- Kaya, O., Rezaei-Tarahomi, A., Ceylan, H., Gopalakrishnan, K., Kim, S., Brill, D.R., 2018. Neural network-based multiple-slab response models for top-down cracking mode in airfield pavement design. *J. Transport. Eng. Part B: Pavements* 144 (2), 04018009.
- Kröger, W., 2008. Critical infrastructures at risk: a need for a new conceptual approach and extended analytical tools. *Reliab Eng Syst Saf* 93 (12), 1781–1787.
- Kröger, W., Zio, E., 2011. *Vulnerable systems*. Springer, London.
- Liu, Y., Zhao, C., Wang, X., Huang, Q., Zhang, X., Yi, D., 2016. The degree-related clustering coefficient and its application to link prediction. *Physica A: Statist. Mech. Its Appl.* 454, 24–33.
- Madni, A.M., Jackson, S., 2009. Towards a conceptual framework for resilience engineering. *IEEE Syst. J.* 3, 181–191.
- Masucci, A.P., Smith, D., Crooks, A., Batty, M., 2009. Random planar graphs and the London street network. *Eur. Phys. J. B-Condens. Matt. Complex Syst.* 71 (2), 259–271.
- Mostafavi, A., Ganapati, N.E., Nazarnia, H., Pradhananga, N., Khanal, R., 2017. Adaptive capacity under chronic stressors: assessment of water infrastructure resilience in 2015 Nepalese earthquake using a system approach. *Nat. Hazards Rev.* 19 (1), 05017006.
- Nazarnia, H., Mostafavi, A., Pradhananga, N., Ganapati, E., Khanal, R., 2016. Assessment of infrastructure resilience in developing countries: a case study of water infrastructure in the 2015 Nepalese earthquake. *Transforming the Future of Infrastructure through Smarter Information: Proceedings of the International Conference on Smart Infrastructure and Construction*, 27–29 June 2016, 627–632.
- Nazarnia, H., Sarmasti, H., 2018. Characterizing infrastructure resilience in disasters using dynamic network analysis of consumers' service disruption patterns. *Civil Eng. J.* 4 (10), 2356–2372.
- Post Disaster Need Assessment (PDNA), 2015. Government of Nepal National Planning Commission. Available online: < <http://www.worldbank.org/content/dam/Worldbank/document/SAR/nepal/PDNA%20Volume%20A%20Final.pdf> > (Accessed in August 2015).
- Porta, S., Crucitti, P., Latora, V., 2006. The network analysis of urban streets: a dual approach. *Physica A: Stat. Mech. Appl.* 369 (2), 853–866.
- Procyk, A., Dhariwal, R., 2010. Transportation sector case study: characterizing vulnerability to infrastructure failure interdependencies (IFIs) from flood and earthquake hazards.
- Proposed Loan Nepal: Urban Water Supply and Sanitation (Sector) Project Report and Recommendation of the President to the Board of Direction September 2018 retrieved January 2019 from: < <https://www.adb.org/sites/default/files/project-documents/35173/35173-015-rrp-en.pdf> > .
- Rasoulkhani, K., Reyes, M.P., Mostafavi, A., 2017. Emergence of resilience from infrastructure dynamics: a simulation framework for theory building. *Comput. Civil Eng.* 2017 (pp. 256–264).
- Reggiani, A., Nijkamp, P., Lanzi, D., 2015. Transport resilience and vulnerability: the role of connectivity. *Transport. Res. Part A: Pol. Pract.* 81, 4–15.
- Robert, B., Senay, M.H., Plamondon, M.E.P., Sabourin, J.P., 2002. Characterization and ranking of links connecting life support networks.
- Winkler, J., Dueñas-Osorio, L., Stein, R., Subramanian, D., 2011. Interface network models for complex urban infrastructure systems. *J. Infrastruct. Syst.* 17 (4), 138–150.
- Yazdani, A., Jeffrey, P., 2011. Applying network theory to quantify the redundancy and structural robustness of water distribution systems. *J. Water Resour. Plan. Manage.* 138 (2), 153–161.
- Yazdani, A., Jeffrey, P., 2012. Water distribution system vulnerability analysis using weighted and directed network models. *Water Resour. Res.* 48 (6).
- Zhao, K., Kumar, A., Harrison, T.P., Yen, J., 2011. Analyzing the resilience of complex supply network topologies against random and targeted disruptions. *IEEE Syst. J.* 5 (1), 28–39.
- Zio, E., Sansavini, G., Maja, R., Marchionni, G., 2008. An analytical approach to the safety of road networks. *Int. J. Reliab. Qual. Saf.* 15 (01), 67–76.